Phase 1, Module 5: Advanced Data Structures

Khyber AI Systems Engineer (KASE)

Welcome to Module 5. As an AI engineer, you don't process one number at a time; you process millions. Today, we learn the four primary architectures for organizing mass data in Python: Lists, Tuples, Sets, and Dictionaries.

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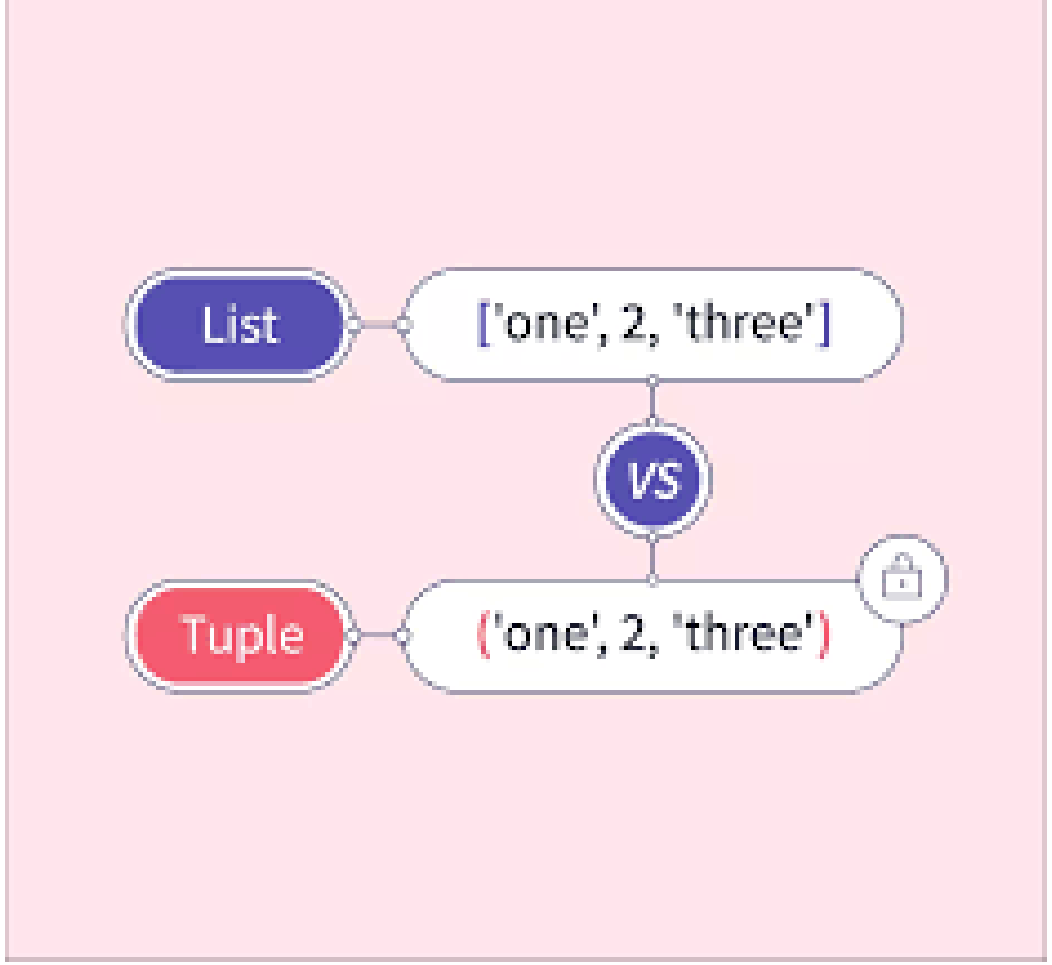
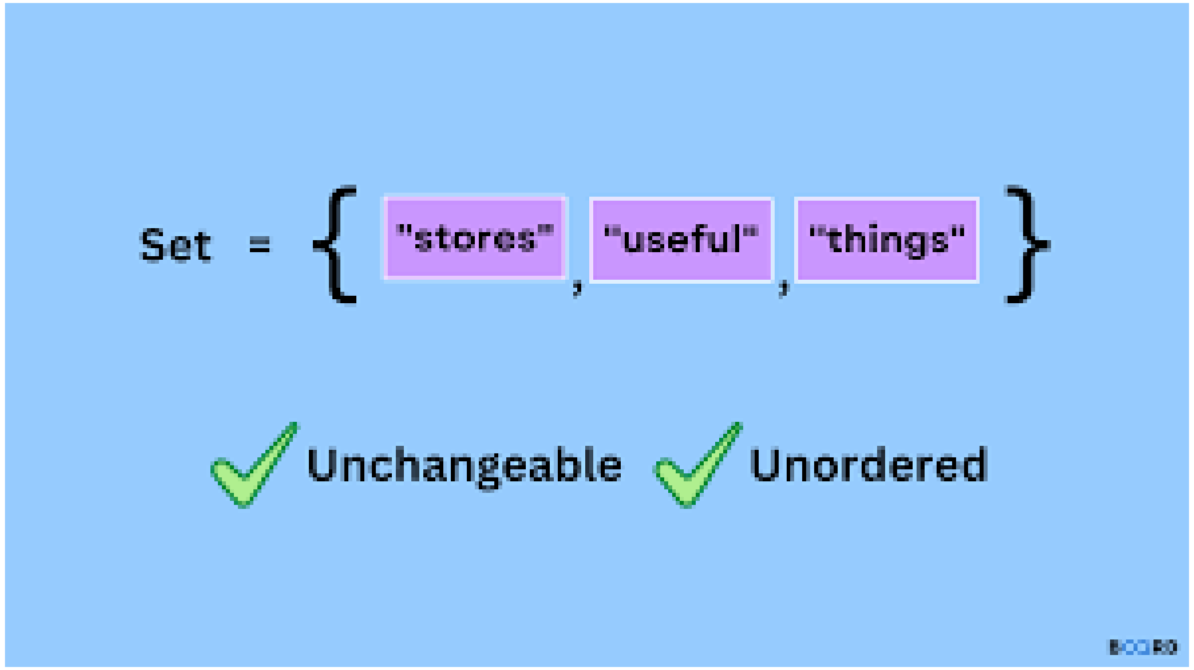


Lists vs. Tuples (The Memory Tradeoff)

Mutability vs. Efficiency

- Lists []: Ordered and Mutable (changeable). Great for dynamic data that grows (e.g., a timeline of temperature readings in Swat).
- Tuples (): Ordered and Immutable.
- The AI Rule: Tuples use less RAM and process faster. We use Tuples for fixed AI configurations, like the shape of an image (224, 224, 3).

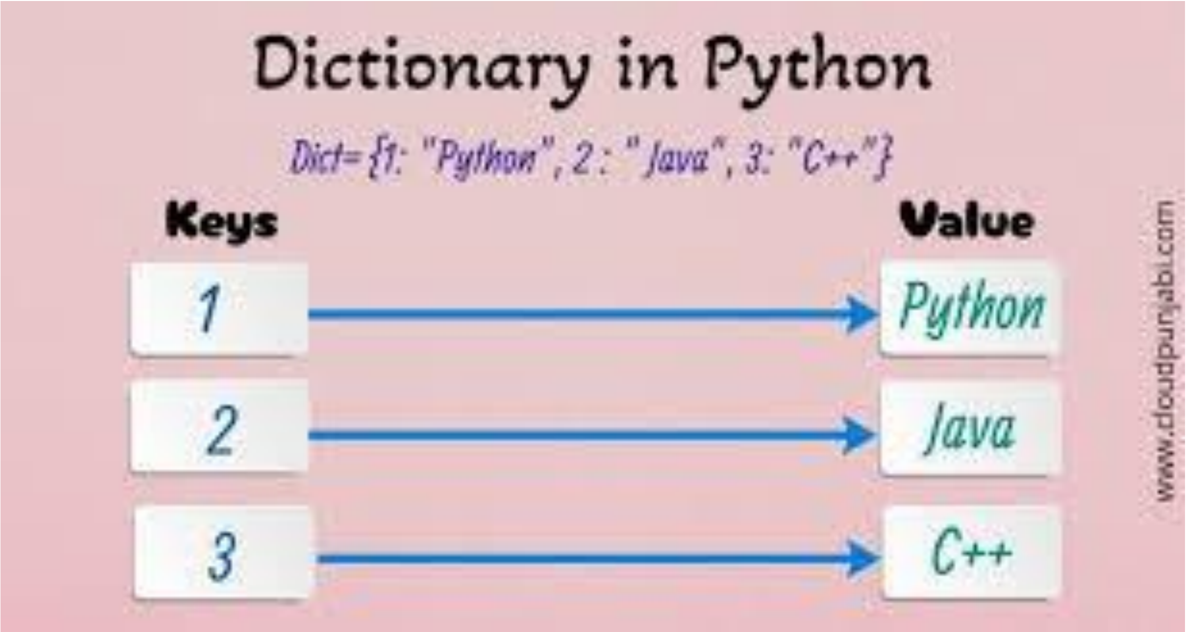
Sets (The Data Cleaner)



Instant Lookups & Deduplication

- **Sets {}:** Unordered, unindexed, and strictly Unique.
- Powered by Hash Tables $O(1)$ time complexity).
- AI Use Case: Cleaning massive, messy datasets.

Dictionaries (The Language of the Web)



Key-Value Mapping & JSON

- Dicts $\{\text{"key"}:\ value\}$: Unordered, mutable, indexed by keys.
- The Golden Rule: Python Dictionaries map perfectly to JSON (JavaScript Object Notation).
- AI Use Case: APIs, Model Hyperparameters, and Agentic Handoffs.